Incident Response & Log Analysis Lab

Objective

The objective of this lab was to analyze failed login attempts using Windows Security logs and identify potential unauthorized access activity.

Tool Used

Windows Event Viewer

Steps Performed

Generated failed login attempts using incorrect passwords

Opened Event Viewer and navigated to Security logs

Filtered logs using Event ID 4625

Reviewed log details such as account name, logon type, and failure reason

Findings

Total failed attempts: 7

Account Name: user1

Logon Type: 2 (Interactive)

Failure Reason: Incorrect password

Time range: 6:48:41 PM – 6:48:53 PM

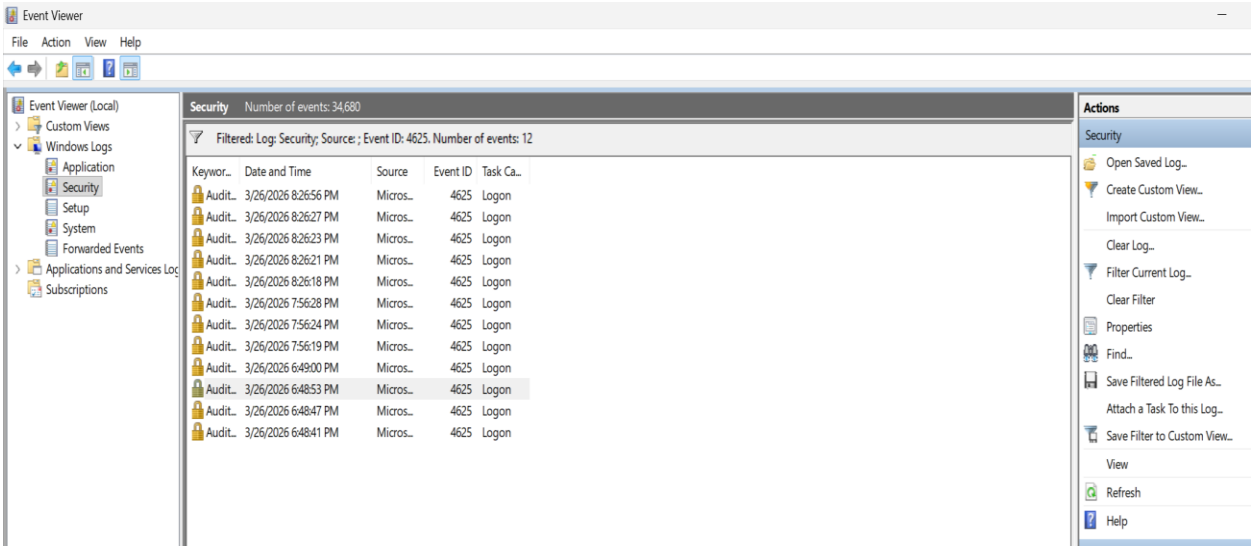

Figure 1: Filtered Windows Security Log displaying multiple failed login attempts (Event ID 4625) for a single user account within a short time frame.

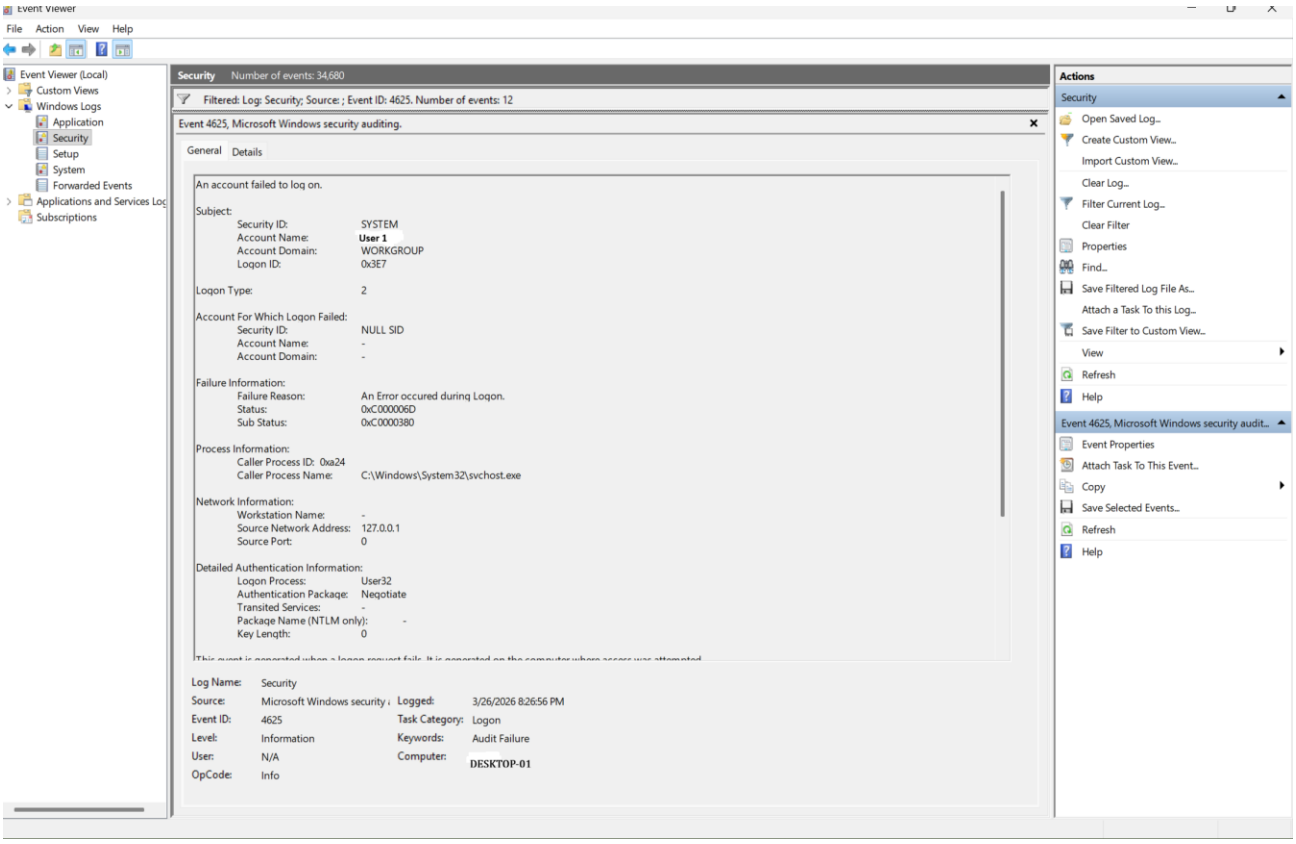

Figure2: Detailed view of a failed login event showing account name, logon type (Type 2 – interactive), failure reason, and timestamp.

Analysis

Multiple failed login attempts were observed for the local user account within a very short time frame. The attempts occurred seconds apart (6:48:41 PM, 6:48:47 PM, and 6:48:53 PM), all targeting the same user account with the same logon type (Logon Type 2 – interactive) and identical failure reason indicating incorrect password entry.

This pattern suggests repeated authentication attempts rather than a one-time mistake. While this could be due to user error, the consistent timing and repetition may also indicate a potential brute-force attempt or unauthorized access activity. If observed in a real-world environment, this behavior would warrant further investigation and possible security controls such as account lockout policies or alerting mechanisms.

NIST Mapping

AC-7 (Account Lockout) → Helps detect and limit repeated failed login attempts

AU-2 (Audit Events) → Ensures logging of authentication activity

AU-6 (Audit Review, Analysis, and Reporting) → Supports analysis of security logs to identify suspicious patterns

SI-4 (System Monitoring) → Enables monitoring for potential unauthorized access attempts

IR-4 (Incident Handling) → Guides response to suspected security incidents

Recommendations

Implement account lockout policies after multiple failed login attempts

Enable alerting for repeated authentication failures

Review logs regularly for suspicious patterns